neovim
Home: https://neovim.io/
Documentation: https://neovim.io/doc/

neovim

neovim 0.5
New features: https://lwn.net/Articles/864712/
A major update with new features.
Potentially breaking features

Features i.e. advantage over vim
Features: https://github.com/neovim/neovim#features
See :help nvim-features
- **asynchronous** job control
e.g. syntax features can be very slow on a large file in classic vim due to synchronous jobs
  syntax highlighting
  syntax checks like Syntastic

- [>=0.5] **Tree-sitter**:
Ref: https://github.com/tree-sitter/tree-sitter
original vim uses **regular expressions** for **syntax**-related features.
e.g. syntax highlighting
e.g. folding
But regex is very limited.
A better approach is to use a real **parser.**
Tree-sitter is a generic parser generator for this.
Neovim uses parsers generated by tree-sitter to perform syntax features.
In order to used syntax features on a **language**, a **language grammar** must be defined for that language so that tree-sitter can generate a **language parser.**
i.e. Similar to how vim required a regex **syntax file** for each language.
Languages are managed by **nvim-treesitter plugin**.

- [>=0.5] **Lua integration**: another step away from **vimscript**
In particular, use a more **generic** language
vimscript is limited to vim, hence not very well designed.
e.g. use lua to configure a plugin

- [>=0.5] **LSP** client:
LSP (language server protocol)
Supports semantic analysis of code
e.g. go to definition
See **neovim LSP** notes for details

LSP vs T**reesitter**
LSP is **semantic**
Treesitter is **syntactic**

Files
~/.config/nvim
~/.config/nvim/init.vim

Help
```
:help nvim
:help nvim-features    # full list of nvim features
:help nvim-from-vim    # Transitioning from vim
```

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Migration from vim

Check if nvim or vim
FAQ: https://neovim.io/doc/user/nvim.html#nvim-from-vim
nvim-from-vim: https://neovim.io/doc/user/nvim.html#nvim-from-vim
Check 'nvim' feature

```
# Help
:help nvim-from-vim

# Check
if has('nvim')
    ...
endif
```

Use python plugins
See **:help provider-python**
Need to install pynvim

Plugins:
YouCompleteMe: python2
Step 1) Install pynvim

```
# Support python2 plugin
pip2 install --upgrade pynvim

# Support python3 plugin
pip3 install --upgrade pynvim
```

Step 2) Check nvim recognizes python

```
" Run
:checkhealth
```

```
" Success
  - OK: Latest pynvim is installed.

" Error
  - ERROR: Python provider error:
```